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A Comparative Analysis of Machine Learning and Deep Learning Approaches for Multiclass Nucleus Classification in Histological Images

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ABSTRACT

Precisely classifying cells in histological images is critical for early cancer diagnosis and tumor assessment. Traditional manual methods are time-consuming and labor-intensive for histopathologists, driving the development of automated approaches using machine learning (ML) and deep learning (DL). Convolutional neural networks (CNNs) and, more recently, vision transformers (ViTs) have demonstrated significant potential in addressing the challenges of cell classification by leveraging their ability to automatically extract and learn complex features from histological images. In this work, we evaluate multiple classification architectures applied to stained histological images to determine their effectiveness in identifying cancerous cells. We compare traditional ML models, which rely on manually extracted features such as shape and texture, against two DL-based classifiers: a CNN-based model (ResNet50) and a ViT-based model. To optimize ML models, we apply principal component analysis (PCA) to refine feature selection. Meanwhile, DL models are trained on cropped cell images using two preprocessing strategies: one that includes additional surrounding cellular context and another that uses only the cell pixels. Additionally, we investigate class balancing strategies, including downsampling and oversampling through data augmentation, to mitigate the effects of dataset imbalance. Experimental results highlight the clear advantage of DL models over traditional ML approaches. ResNet50 consistently delivers robust and reliable performance across different preprocessing strategies, confirming its effectiveness for histopathological classification tasks. Meanwhile, ViTs achieve results that are comparable to those of CNNs while demonstrating a distinct advantage in classifying underrepresented nucleus classes, likely due to their ability to capture long-range dependencies. Furthermore, incorporating the surrounding cellular environment significantly improves classification accuracy, underscoring the importance of contextual information in distinguishing between different types of nuclei.

1 | Introduction

According to statistics presented by Bray et al. [1], in 2022, cancer was responsible for nearly 10 million deaths worldwide, with close to 20 million new cases reported. They also suggest that one in five people could develop cancer at some point in their lifetime. Additionally, they predict that these numbers will increase significantly due to population growth and aging, with an estimated 35 million new cases annually by 2050. This alarming trend highlights the urgent need for advancements in early detection, precise diagnostic tools, and effective treatment strategies to address the growing burden of this disease.

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Early cancer diagnosis can significantly improve survival rates, according to Hawkes [2]. However, as of today, no universally accepted method for cancer diagnosis exists. While several promising techniques are emerging, histopathological imaging remains the gold standard for diagnosing many types of cancer [3]. Traditionally, pathologists rely on microscopy to examine hematoxylin and eosin (H&E)–stained images. This process is time-consuming, highly subjective, and requires years of specialized training and experience to provide accurate diagnoses [4]. Additionally, human error and inconsistencies in image interpretation remain significant challenges, underscoring the need for more reliable and automated diagnostic methods.

Among other methods, accurately classifying cells in histological images can assist early and effective cancer diagnosis. Indeed, precise classification of cells, along with their number and size, can be determining factors in assessing tumors' aggressiveness.

Computational pathology (CPath) solves these challenges by enabling a faster and more accurate analysis of digitized slides or whole-slide images (WSI). CPath can extract meaningful features from these images through computational analysis, identify the region of interest (ROI), and segment or classify the tissue. It can significantly reduce the workload on pathologists while enhancing diagnostic accuracy. For using CPath in tasks such as diagnosis and classification, a typical workflow [4] consists of the following: (1) digitization of glass slides to create WSI; (2) image preprocessing, such as stain normalization, morphological operations, tissue segmentation, and tiling; (3) deep learning pipelines involving data splitting, feature extraction, model training with hyperparameter optimization, and evaluation in validation and test sets; and (4) application of trained models for cancer diagnosis, classification, prognosis prediction, and quantitative evaluation of the tumor microenvironment.

For image analysis in CPath, deep learning models are widely used. For example, convolutional neural networks (CNNs) and their variations have shown their effectiveness in handling complex image data [5]. On the other hand, recently, vision transformers (ViTs) [6] have also gained attention for their promising performance in image analytics. All these models have been successfully applied in CPath for tasks such as segmentation and classification, demonstrating their potential to improve diagnostic tools [5, 7, 8].

In this work, among all the possibilities of CPath, we are interested in cell classification tasks where the goal is to analyze images at the cellular level to extract information regarding structure, function, or pathology.

Note that, traditionally, cell classification tasks rely on features such as cell morphology, texture, or intensity, often in combination with traditional machine learning techniques [9–11]. In recent years, CNNs have become more prevalent, and even transformer-based architectures, such as ViT, have gained popularity for these tasks.

In this work, we aim to apply CPath to the cell classification of H&E-stained images by using the PanNuke dataset [12, 13]. First, to have a baseline analysis, we study several machine learning algorithms and evaluate their capabilities and performance. To train these algorithms, we use the most significant features chosen from the analysis of several studies. Second, we study the behavior of deep learning models by using a CNN model and a transformer model. For both, we analyze two different preprocessing pipelines. Finally, we provide a comparative analysis of the behavior of these models with the behavior of traditional ML models. This comparison allows us to evaluate their capabilities and performance.

Our first goal focuses on feature extraction and data analysis using the PanNuke dataset. We investigate key features for cell classification, implement feature extraction code, and conduct an exploratory data analysis (EDA) to understand the relationships between features and cell classes. Additionally, we perform a principal component analysis (PCA) to reduce the dimensionality of the features. Our second goal is to implement several machine learning models, such as random forest, K-nearest neighbors (k-NN), logistic regression, and gradient boosting. These algorithms use the previously extracted features. Our third goal is to implement a residual network (ResNet) 50 model and a ViT; these models directly process cell images without extracting the features. Finally, our last goal is to evaluate and compare the performance of the implemented models.

Our results show that deep learning models, particularly ResNet50 and ViTs, achieve significantly better performance compared to traditional machine learning algorithms. ResNet50 demonstrates consistently better performance across several preprocessing strategies. ViTs outperform ResNet50 in classifying some underrepresented nuclei classes. Furthermore, our findings emphasize the importance of incorporating cell context into classification tasks. By including the surrounding cellular environment as part of the input data, we observe improvements in classification accuracy, revealing the critical role of contextual information in distinguishing between different nuclei types.

2 | Related Work

Over the years, various methods have been developed for mitosis classification, generally falling into two main categories: strategies based on manual feature extraction and methods based on deep learning [14]. A third category, often recognized, combines elements of both approaches [10].

2.1 | Manual Feature Extraction Methods

Manual feature extraction methods rely on handcrafted features to capture the relevant morphological, textural, and color-based characteristics of mitotic figures. These methods typically involve a sequence of image preprocessing, feature computation, and classification using traditional machine learning methods such as support vector machines (SVMs), random forests, or k-NN [10, 14]. These methods have been used for the task of cell classification [11, 15]. The quality of the selected features limits the capability of these algorithms.

2.2 | Deep Learning Methods

Deep learning methods have revolutionized mitosis classification by leveraging CNNs and other deep architectures to automatically learn hierarchical features from data. Unlike manual feature extraction approaches, DL models do not require predefined features but instead optimize feature representations directly from raw image data [10, 14]. Several research studies [3, 5, 16] confirm that CNN architectures provide better performance,

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specifically ResNet50, which is the most commonly used for this task. Moreover, the ViT has recently emerged as a powerful alternative.

2.3 | Literature Review

The PanNuke dataset [13] has become a widely adopted benchmark for nuclei image classification in CPath due to its large scale, multitissue diversity, and detailed annotations. It contains more than 200,000 nuclei from 19 tissue types, each labeled into five morphological classes: neoplastic, inflammatory, connective or soft tissue, dead, and epithelial. Most classification studies on PanNuke focus on directly discriminating these nucleus categories from image patches or as part of multitask pipelines that combine segmentation and classification.

Prior to the dominance of deep learning, nuclei classification was usually approached through handcrafted features and classical machine learning classifiers, but such methods have seldom been applied to PanNuke. A notable example is the study by Yellu et al. [17], which employed handcrafted morphological and texture features from nuclei patches, combined with ensemble classifiers, to perform classification on the PanNuke dataset. Their results achieved accuracies around 78% to 82% for dominant classes, demonstrating that although deep learning remains superior, classical machine learning can still provide interpretable and competitive baselines for nuclei classification tasks.

Early work by Gamper et al. [18] introduced the extended PanNuke dataset and provided the first classification benchmarks using CNNs trained on cropped nucleus patches. Their results demonstrated the feasibility of deep learning for nuclei classification, reporting average $F1$-scores around 0.80 for dominant classes and highlighting the challenges posed by class imbalance, particularly for rare categories such as dead and connective nuclei. Building upon this, Graham et al. [19] proposed a HoVer-Net, a multibranch CNN capable of simultaneous nucleus segmentation and classification. When evaluated on PanNuke, the HoVer-Net achieved the best results, demonstrating that multitask architectures can leverage shared representations to improve classification accuracy compared to classification-only CNNs.

More recent studies have explored architectural enhancements to improve classification performance. Chen et al. [20] introduced HistoNeXt, a ConvNeXt-based CNN that uses dual-mechanism feature fusion to strengthen representation learning. On PanNuke, HistoNeXt achieved an overall classification $F1$-score of 0.82, with class-specific scores ranging from 0.71 for neoplastic nuclei to 0.36 for dead cells. Transformer-based architectures have further advanced performance: Hörst et al. [21] proposed CellViT, a ViT that achieved a detection $F1$-score of approximately 0.83 on PanNuke, while Pina et al. [22] introduced CellNuc-DETR, a DETR-style detection-classification model that surpassed segmentation-based methods such as HoVer-Net in classification performance and improved inference speed.

Alternative deep learning methodologies have also been explored. Yao et al. [23] presented PointNu-Net, a keypoint-based model that identifies nucleus centroids before performing classification, achieving improved classification metrics compared to conventional CNNs. Chen et al. [24] developed TSHVNet, an enhanced version of the HoVer-Net that incorporates scale and texture modules, improving classification $F1$-scores by 2%-3% over the original architecture.

Many prior studies have combined nuclei detection or segmentation with classification, aiming to localize nuclei in histopathological images and then assign each detected nucleus to its specific cell type. These joint detection-classification approaches have the advantage of leveraging spatial and morphological context during segmentation, which can boost classification accuracy. In contrast, Sun et al. [25] shifted the focus away from detection and instead concentrated exclusively on nuclei classification. They applied classification models across multiple datasets—rather than committing to a particular segmentation pipeline—and systematically compared the performance of different architectures and strategies for pure classification tasks.

Our study presents a comparative evaluation of machine learning and deep learning approaches for the multiclass nuclei classification in the PanNuke dataset. Unlike previous works that focused exclusively on handcrafted features or deep segmentation–classification pipelines, our research systematically contrasts traditional algorithms—trained on morphological, intensity, and texture-based features—with end-to-end models, such as ResNet50 and ViTs, trained directly on nuclei images. Two preprocessing strategies, with and without surrounding cellular context, and class-balancing techniques are examined to assess their influence on performance.

3 | Materials and Methods

Figure 1 explains the workflow of this study. First, it consists of feature analysis of cells and then training several traditional machine learning algorithms and some deep learning algorithms. The trained models are then used to classify the cells in the PanNuke dataset.

3.1 | Dataset

For cell classification datasets, we utilize the PanNuke dataset [13, 18]. This dataset is a public database for research purposes only.

The PanNuke dataset is employed to classify nuclei into five categories: neoplastic, non-neoplastic epithelial, inflammatory, connective tissue, and dead cells. This dataset was developed using a semiautomatic annotation pipeline combining expert pathologists' input and computational methods, which include the following steps [13]:

- Aggregation of nuclei detection datasets, including Kumar, CPM17, and other sources such as The Cancer Genome Atlas and bone marrow fields.
- Training of a CNN for nuclei classification, followed by expert validation and reannotation.
- Integration of 2000 visual fields from more than 20,000 WSIs spanning 19 tissue types, ensuring diversity and the inclusion of clinical artifacts.
- Use the NuClick method for efficient segmentation mask generation based on point annotations.

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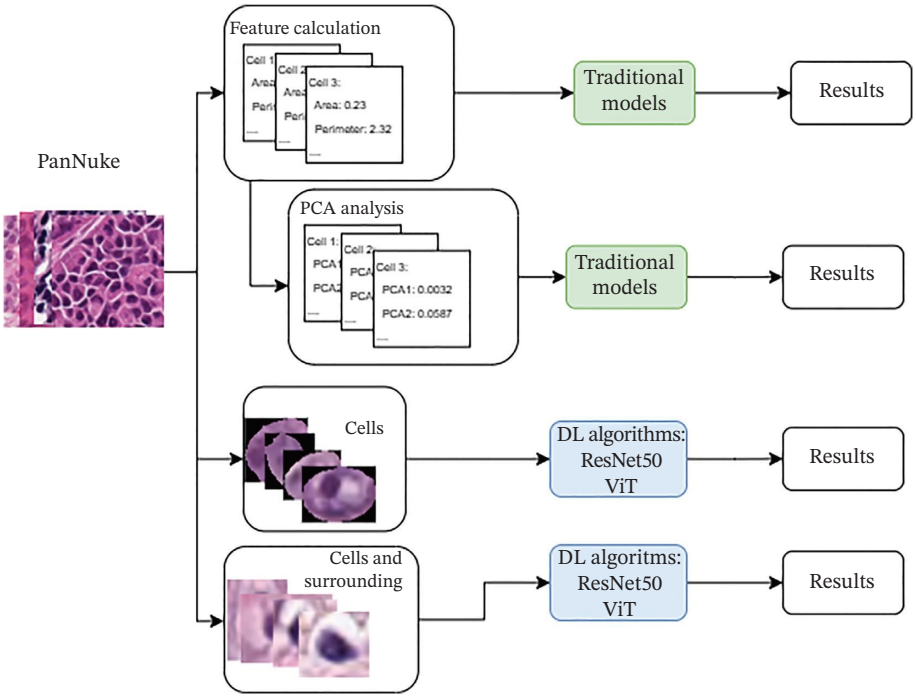

FIGURE 1 | Overview of experimental pipeline using the PanNuke dataset. The pipeline consists of a comparison between traditional feature-based models (feature calculation, PCA) and deep learning approaches (ResNet50, ViT) applied to isolated cells and cell surroundings.

The PanNuke dataset comprises 481 visual fields, each with 189,744 nuclei exhaustively annotated and categorized. This dataset provides a robust schema for nuclei classification in a wide range of tissues and clinical contexts. The dataset is divided into three folds and is commonly used for training, validation, and testing. We download the dataset from the Warwick University webpage (https://warwick.ac.uk/fac/cross_fac/tia/data/pannuke).

PanNuke is a highly unbalanced dataset, as it contains 77,403 neoplastic nuclei, 50,585 connective nuclei, 32,276 inflammatory nuclei, 26,572 non-neoplastic epithelial nuclei, and only 2908 dead nuclei. This imbalance may pose challenges for accurately classifying the cells.

3.2 | Class Balancing

To address the class imbalance inherent in the PanNuke dataset, we apply two complementary strategies. In the first approach, we employ undersampling or downsampling to create a balanced subset, where each cell class contains an equal number of samples corresponding to the least represented class, identified as dead cells. Using the first fold of the PanNuke dataset for training, we randomly select approximately 900 samples per cell type to achieve uniform representation across all classes. Secondly, as an alternative balancing strategy, we apply oversampling through data augmentation to increase the number of samples in the minority classes until their representation matches that of the most prevalent class, neoplastic cells, which contain approximately 26,000 samples in the first PanNuke fold. Augmentation techniques include a combination of geometric transformations, such as rotations, flips, and their combinations, as well as color-based augmentations, including slight variations in brightness, contrast adjustments, mild Gaussian noise, small blurring effects, and minor alterations in color channels. When necessary, geometric and color transformations are applied jointly to enhance variability. In total, 22 distinct transformations are generated, complemented by 5 additional combined geometric and color transformations, resulting in a final balanced dataset consisting of approximately 26,000 samples per class.

3.3 | Feature Selection, Calculation, and Analysis

To implement the traditional ML algorithms, we conduct a literature review to identify the most widely used features. We analyze five different studies [9–11, 15, 26].

3.3.1 | Feature Selection and Calculation

We select features reported in at least two of the reviewed studies. Note that some features are similar but not identical. For example, Wang et al. [10] employ, as texture features, the same 13 Haralick properties [27] employed by Amgad et al. [11]. However, the first calculates the mean and standard deviation, while the second uses the mean and range. We decided to use only the mean for these 13 textural features.

Regarding the intensity or color-based features, some papers use RGB color and others use YUV, but we rely on the hematoxylin channel [11].

Table 1 shows the feature categories analyzed, and the last column shows the proposed categories. Table 2 lists the selected features for each category.

The feature calculation process analyzes the PanNuke dataset, extracts cells using a mask, and calculates various features. For shape features, it uses OpenCV [28] to detect contours and discards cells without enough data, such as those at the edges. It converts the image to a specific channel for intensity features and measures values such as minimum and median. Finally, for

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TABLE 1 | Feature categories and normalization.

Kode et al. [9]	Wang et al. [10]	Agmad et al. [11]	Aswathy and Jagannath [15]	Unified categories
Geometrical	Morphology	Size Shape	Geometrical	Morphology
Directional		Edges		—
Intensity-based	Intensity-based	Intensity-based	Color	Intensity
	Textural	Textural	Textural	Textural

Note: The last column shows the proposed categories. We do not include Corredor et al. [26] since they do not provide any categories.

texture features, it uses the Mahotas library to compute 13 texture-related characteristics.

3.3.2 | Analysis of Calculated Features

To better understand the dataset and the features extracted, we analyze the first fold of the PanNuke dataset, which is used for training. Our initial step is to calculate the correlation matrix of the features to analyze correlations and remove highly correlated features.

The feature correlation analysis revealed that features within the same category, such as texture, intensity, or morphological features, tend to be highly correlated, indicating that they share similar patterns or provide overlapping information. For example, there is a strong correlation between the features "Area" and "Perimeter," and there are high correlations between the morphological features and the Haralick texture features.

Additionally, some features are completely correlated, and their values can be perfectly predicted based on other features. These features do not contribute unique information and are considered redundant [29]. They do not add value and might even introduce noise or unnecessary complexity to the model. Consequently, we exclude angular second moment (ASM), inverse difference moment (IDM), entropy, and sum entropy from training to ensure that the model focuses only on features that provide distinct, meaningful information. This step not only improves model efficiency and performance but also reduces training time by decreasing the number of features used.

We analyzed the distribution of each feature by category. As an example, Figure 2 shows the equivalent diameter feature that exhibits a normal distribution in general and for each cell type.

Figure 2 shows that neoplastic cells tend to have a higher equivalent diameter, with a mean around 25, while dead cells have a mean close to 10. This highlights how the equivalent diameter can differentiate between these two cell types. Additionally, the epithelial cell type is the most similar to neoplastic in terms of equivalent diameter. This similarity underscores the challenge of distinguishing between these two cell types.

We analyze all features separately. This analysis shows us that features such as area, major axis, and minor axis, among others, have noticeably different mean values across the categories.

3.3.3 | PCA

We perform a PCA to evaluate how the principal components impact the performance of our trained models. The PCA reduces the dimensionality of the dataset while retaining most of the critical information. It allows us to maximize machine learning workflows.

We calculate the variance ratio of each PCA component and its cumulative variance. This helps us to determine how many components are necessary to represent the data effectively. The analysis shows that using between 10 and 15 components retains 95% to 99% of the total explained variance. Therefore, a significant portion of the information in the original dataset can be captured with far fewer features.

Reducing the number of components decreases the model's complexity. It also reduces the risk of overfitting and speeds up the training process. Figure 3 shows the cumulative importance of aggregating the components, illustrating how the cumulative explained variance increases as additional components are added.

3.4 | Traditional ML Models

The use of conventional ML algorithms is based on their ability to provide a solid baseline for the problem. In this study, we select random forest, k-NN, logistic regression, and gradient boosting. These algorithms represent diverse approaches that capture the

TABLE 2 | List of features selected by category with its total number of features.

Category	Length	Features
Morphology	8	Area, perimeter, circularity, equivalent diameter, solidity, major axis, minor axis, extent
Intensity-color	9	Minimum, maximum, mean, median, standard deviation, range, interquartile range, skewness, kurtosis (for hematoxylin channel)
Texture	13	Angular second moment, contrast, correlation, sum of squares, inverse difference moment, sum average, sum variance, sum entropy, entropy, difference variance, difference entropy, information measures of Correlation 1, information measures of Correlation 2 (mean values)

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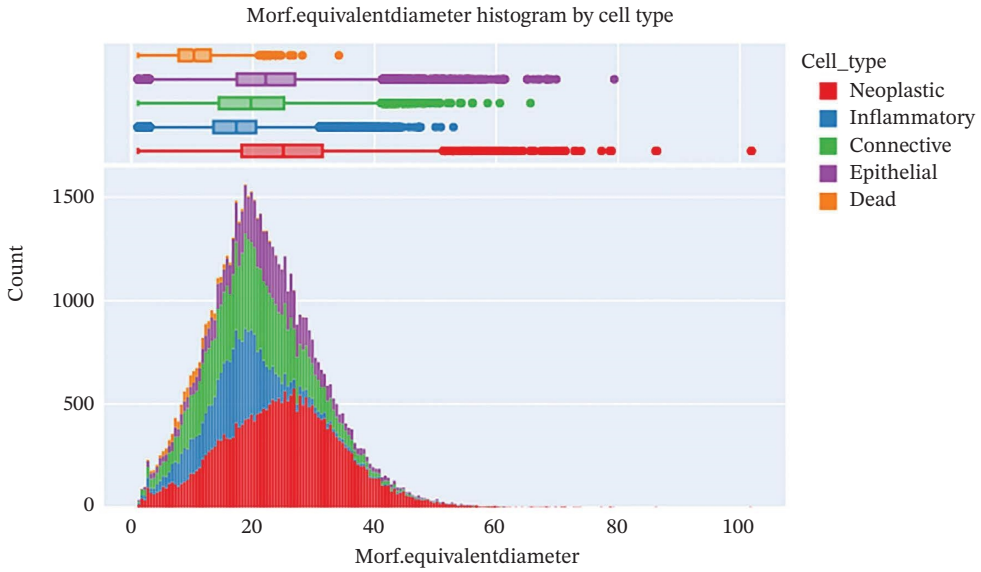

FIGURE 2 | Distribution of equivalent cell diameter grouped by cell type. The histogram shows the frequency of values, while the boxplots highlight variability and outliers within each group. Neoplastic cells tend to have a higher equivalent diameter, with a mean around 25, while dead cells have a mean close to 10.

data's linear, nonlinear, and local relationships. This initial analysis establishes a reliable foundation for comparison with the performance of more advanced models such as ViT and CNNs. This comparison allows for measuring their added value in accuracy and predictive capacity.

We extract the cells and calculate all the features for each part of the dataset: training, validation, and test sets. These calculated features are the basis for our traditional models to generate results.

We utilize the *scikit-learn* library to train the random forest, k-NN, logistic regression, and gradient boosting algorithms using the extracted features.

We split the PanNuke dataset into three folds, based on the existing dataset structure. These algorithms are trained with the first fold of PanNuke, using a Grid of hyperparameters for each one. After that, we used the third fold to test the best-trained model. In this way, we get an overview of how each algorithm

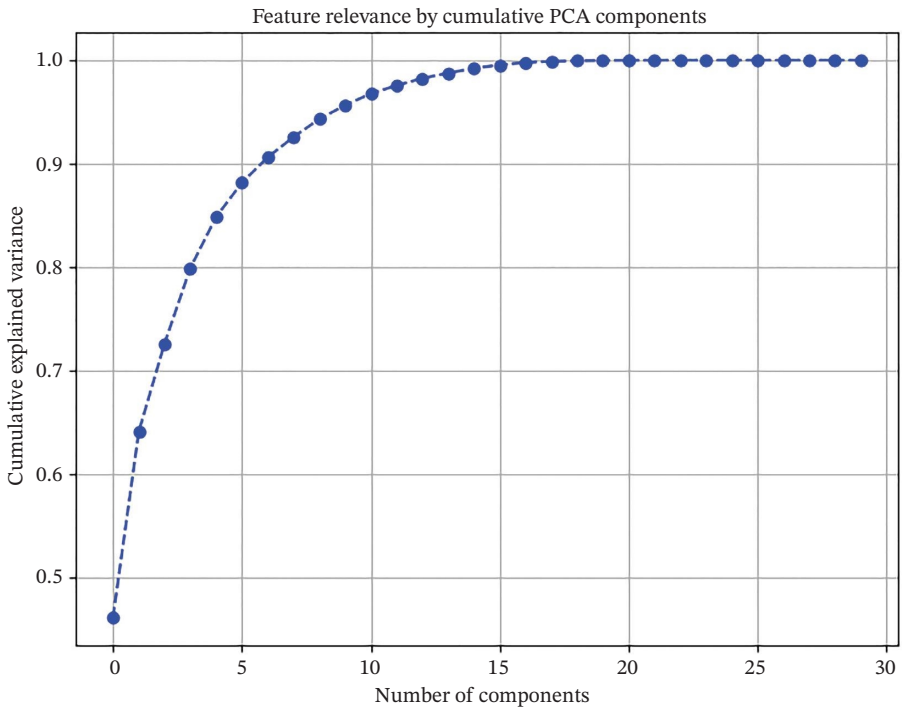

FIGURE 3 | Cumulative explained variance as a function of the number of PCA components, showing the proportion of total variance retained by increasing component inclusion. The cumulative explained variance increases as additional components are added.

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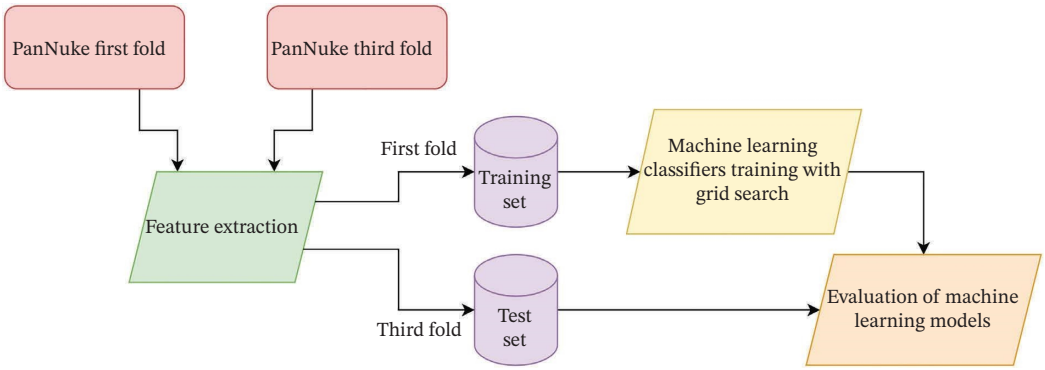

FIGURE 4 | Workflow of the machine learning classification pipeline from feature extraction on the PanNuke dataset to model training with grid search and final evaluation.

performs. We use the second fold to validate deep learning models. Figure 4 shows a diagram of how the training is used.

3.5 | Deep Learning Algorithms

We select ResNet50 and ViT as deep learning algorithms since both are known for their state-of-the-art performance in computer vision tasks. These DL algorithms are highly effective due to their deep architectures, which allow the models to automatically extract and learn complex features and patterns directly from the input data.

Predefined feature calculations do not limit deep learning models, enabling them to discover intricate and high-dimensional patterns that might be overlooked in manual feature engineering. ResNet50, with its residual connections, excels at addressing the vanishing gradient problem, making it highly efficient for training deep networks. The ViT, on the other hand, leverages self-attention mechanisms to process image data, capturing global dependencies and relationships within the input.

3.5.1 | ResNet50

ResNet, commonly referred to as ResNet, is a type of CNN (cite he2015). It is built as a traditional CNN but adding residual connections that connect a layer not only with the next layer but also with a layer separated by one or more steps. This residual connection allows the information to bypass certain layers, ensuring that the network can learn identity mappings when additional depth does not improve the model's performance. Figure 5 shows, at the bottom, a traditional CNN connection between layers, while at the top it shows the connections of a ResNet with the residual connections.

ResNets are widely known for their ability to train profound networks, sometimes consisting of hundreds or thousands of layers, without suffering from performance degradation. This makes them particularly effective in tasks such as image classification and object detection. In CPath, they are widely used as backbones for classification tasks, often pretrained on ImageNet. Variants of ResNet, such as ResNet-50, ResNet-101, and ResNet-152, are distinguished by the number of layers in the network.

The ResNet50 is initialized with pretrained weights on ImageNet from the torchvision library. This way, we benefit from transfer learning and generalization. We also modify the last classification layer to our desired cell type categories.

3.5.2 | ViTs

ViTs are DL model architectures that apply the principles of transformers. They were initially developed for natural language processing (NLP), but a recent study by Dosovitskiy et al. [6] introduced the application of transformers for image processing. The idea is to divide the image into smaller patches and treat them as a sequence, in the same way they are treated on NLP tasks.

On ViT, images are divided into fixed patches flattened into vectors. They are embedded with positional encodings to preserve spatial relationships and then fed into transformer encoders as shown in Figure 6. The encoder consists of self-attention mechanisms. This enables the model to capture local and global data patterns. Since the ViTs appeared, they have shown competitive performance compared to CNNs.

The ViT model is initialized with pretrained weights from the ImageNet checkpoint, ensuring the model benefits from transfer learning and generalization. By specifying the number of labels equal to the unique cell types (five) in our dataset, we adapt the model to predict our specific classification task.

3.5.3 | Data Preparation for DL Models

To train and validate these two algorithms, we follow a similar route.

- Data Loading and Preparation: The dataset is split into training, validation, and testing subsets using a 3-fold split based on the PanNuke dataset. The first fold is used for training, the second for validation, and the third for testing. The metadata associated with each image is preprocessed, and the labels are encoded to ensure compatibility with the PyTorch framework. Images are resized and padded to ensure uniformity, as shown in Figure 7. The SquarePad transformation ensures that all input images retain their aspect ratios while resizing to fit the 224×224 input size expected by the ViT model and ResNet50. This preprocessing pipeline also includes data augmentation techniques, such as random horizontal or vertical flipping and normalization, which enhance the model's generalization capability. We train the models using two different datasets. One dataset includes

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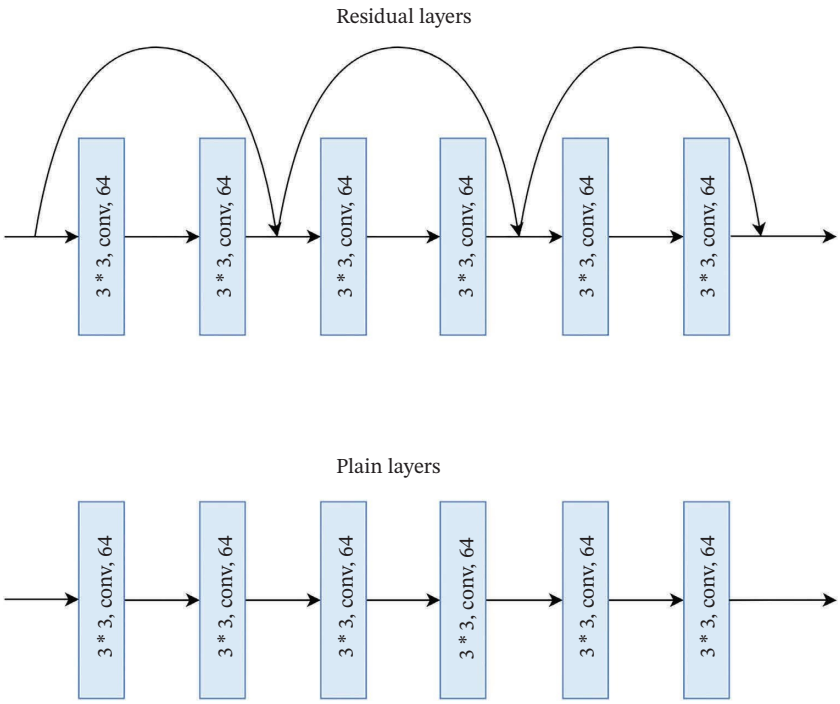

FIGURE 5 | Comparison between residual layers and plain layers. Residual connections, introduced in ResNet [30], enable deeper training by facilitating gradient flow.

a 10-pixel buffer around each cell, providing the models with a small context of the surrounding area (see Figure 7). The other dataset excludes the surrounding areas, using only the cell mask for classification (see Figure 8).

- Optimization and Loss Functions: We utilize the Adam optimizer with a learning rate of 0.001, which strikes a balance between convergence speed and stability. The loss function is cross-entropy loss, suitable for multiclass classification problems.
- Performance Metrics: During training, key metrics, including accuracy, recall, $F1$-score, and loss, are calculated to track the model's performance. These metrics are aggregated per epoch to assess improvements over time.
- Checkpoints: We monitor the validation $F1$-score to prevent overfitting. The best-performing model on validation is saved for evaluation on the test set. The checkpointing mechanism saves the model's weights and metrics at each epoch, ensuring reproducibility.

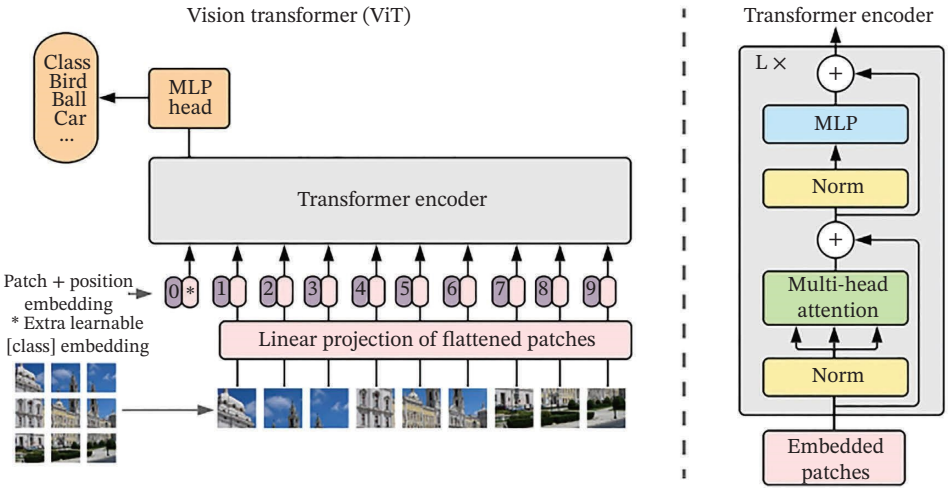

FIGURE 6 | Overview of the ViT architecture, extracted from Dosovitskiy et al.'s study [6]. The model processes an input image as a sequence of patches, which are linearly embedded, combined with positional encodings, and passed through a transformer encoder.

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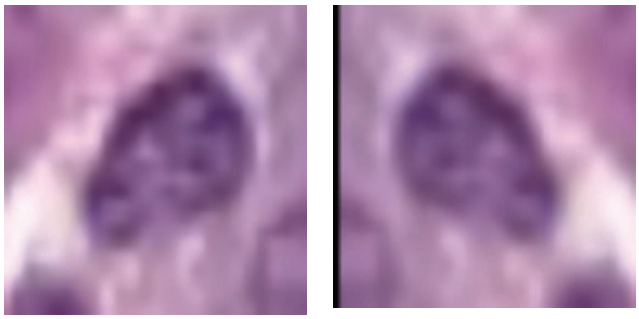


(a) (b)

FIGURE 7 | (a) A neoplastic cell extracted from the first fold of the PanNuke dataset, using the segmentation–classification mask and expanded by 10 pixels in each direction, resulting in a resolution of 47×48 pixels. (b) The same cell after applying transformations to resize it for input into the ResNet50 and vision transformer models, with the image resolution adjusted to 224×224 pixels.

- We load both models onto the GPU device to benefit from the optimized processing capabilities and to reduce training and testing times.

4 | Results and Discussion

For the classification task, we evaluate the results using accuracy and $F1$-score as the metric. We also provide training and test times for each algorithm. For all the experiments, we use the PanNuke dataset.

4.1 | High-Performance Setup

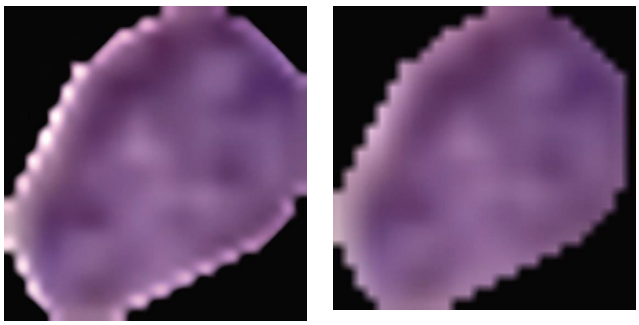
Our experimental platform consists of a system featuring a dual AMD EPYC 9124 CPU and 128 GB DDR4 DRAM. This system features an NVIDIA GeForce RTX 4090 GPU with 24 GB of GDDR6X memory. All experiments are conducted using the GPU, harnessing the computational power of the RTX 4090. This high-performance environment provides excellent performance and fast convergence for our deep learning experiments. We use Python 3.9.7, scikit-learn Version 1.6.0, PyTorch Version 2.5, and transformers Version 4.47.1 for our implementations. All the code developed is available at https://github.com/antonio99luis/ResNetVsTransformers.git, but access is granted upon request via email.

4.2 | Machine Learning Algorithms

First, we utilize ML algorithms not only to gain a broad understanding of the potential but also to establish a baseline classification score. We employ four machine learning models: random forest, k-NN, logistic regression, and gradient boosting. Table 3 presents results for these models based on raw features and the first 11 principal components. Hyperparameter selection for each model is performed using the training data, ensuring the optimal configurations are used.

As observed, random forest performs better when trained on both raw features and principal components. The k-NN algorithm improves the principal components compared to the raw features. This may be attributed to the high dimensionality of the raw features. In high-dimensional spaces, the Euclidean distance metric often loses its discriminative power, making it difficult to differentiate between the nearest and farthest points, a phenomenon known as the curse of dimensionality [31]. The logistic regression also improves its results on principal components because PCA removes the multicollinearity of the features.

In contrast, random forests and gradient boosting can extract meaningful insights directly from the raw features without requiring transformations that might distort the underlying differences.



(a) (b)

FIGURE 8 | (a) A neoplastic cell extracted from the first fold of the PanNuke dataset, using the segmentation–classification mask, in a resolution of 27×28 pixels. (b) The same cell after applying transformations to resize it for input into the ResNet50 and vision transformer models, with the image resolution adjusted to 224×224 pixels.

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TABLE 3 | Classification metrics on test data for the traditional machine learning algorithms trained on the raw features and on the first 11 PCA components.

Model	Raw features		PCA components	
	Acc	*F*1	Acc	*F*1
Random forest	**64.01**	**62.99**	**55.14**	**53.38**
k-NN	52.63	51.06	53.87	52.69
Logistic regression	45.10	35.31	52.58	49.78
Gradient boosting	63.43	62.54	53.37	52.54

Note: Bold values represent the best results achieved in each metric.

Table 4 presents training and test times for each algorithm and data input. The fastest algorithm to train is k-NN, while the slowest is gradient boosting. For prediction, the fastest algorithm is logistic regression, and the slowest is random forest. It is also important to note that some hyperparameters had a significant impact on the algorithms' training. As it is logical to suppose, the times on principal components are lower for training, since they had fewer input parameters for each nucleus.

4.3 | Deep Learning Algorithms

As explained previously, we use ViT and ResNet50 for the DL algorithms, training them on two different preprocessed inputs. One input consists solely of the cell mask, while the other includes a margin of 10 pixels in each direction, allowing the model to capture both the cell and some surrounding context. During training, we test each epoch in the validation set to prevent overfitting, and the best result on the validation set is used to obtain the final testing results. We observe how training and evaluation metrics improve gradually for both models, but toward the final epochs, the validation metrics stabilize. The validation metrics peak at Epoch 20 for ViT and 8 for ResNet50 on this dataset.

For the dataset without surrounding areas, using only the cell, the best results are achieved at Epoch 30 for ViT and 17 for ResNet50.

After training, we test both models using the best epochs in the test set. Table 5 presents the training results for each model in both datasets. As shown, the results are quite similar, demonstrating the capabilities of both models as backbones for more complex, state-of-the-art models in nuclei classification. However, ResNet50 performs slightly better on average with this preprocessing. We also observe that including the surrounding areas of the cell plays a significant role in improving the performance of both algorithms.

We further compare the performance of each algorithm by nuclei class, as shown in Table 6. This table reveals several notable observations. First, it indicates that ResNet50 does not outperform the ViT across all classes. This may be because ResNet50 extracts more valuable information for the classes better represented in the dataset. The ViT also exhibits this bias, even to a slightly lesser extent. Notably, dead cells account for less than 1.5% of the dataset, which likely caused both models to fail to learn important features for these cells. Meanwhile, neoplastic cells, which represent more than 40% of the dataset, achieved the most accurate results, with up to 80% on accuracy and *F*1-score metrics.

However, the results are not entirely aligned with this assumption, as the ViT classifies epithelial nuclei more accurately than inflammatory nuclei, despite the latter being more represented in the dataset. The difference is relatively small compared to the disparity between neoplastic and dead cells. Specifically, inflammatory nuclei have only about 5000 more samples than epithelial nuclei, which constitute approximately 3% of the entire dataset. Another potential reason for the poor performance on dead nuclei could be the reliance on size-related features as key discriminators for this class. These features tend to have lower values for dead cells, but during preprocessing, all nuclei were resized to match the 224×224 pixel input required for both algorithms.

Finally, the processing times for each algorithm are shown in Table 7. We can see here that in this case, the time taken is the same for both preprocessing strategies. It is normal since the input is the same 224×224 array, but with more or fewer black pixels. We can also notice that for both training and testing time, the ResNet50 is faster than the ViT, averaging 2 and a half minutes less per epoch, while for testing, it also outperforms the ViT by half a minute.

4.4 | Class Balancing Results

4.4.1 | Machine Learning Algorithms With Class Balancing

We apply the class balancing strategies described in Section 3.2 to the machine learning models that use handcrafted features. The corresponding results are presented in Table 8. Among all models, random forest continues to demonstrate the best

TABLE 4 | Time results of classification tasks for machine learning algorithms trained on the raw features and in PCA components.

Model	No. of combination of hyperparameters	Raw features		PCA components	
		Training (s)	Test (s)	Training (s)	Test (s)
Random forest	74	302	0.7	136	0.6
k-NN	18	**17**	3.5	**7.3**	9.7
Logistic regression	12	59	**0.004**	29	**0.03**
Gradient boosting	28	2832	0.27	1269	0.48

Note: Bold values represent the best time (lowest time) achieved in each stage.

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TABLE 5 | Classification metrics on test data for the DL algorithms for both preprocessing image strategies.

Model	Nuclei and surroundings		Just nuclei	
	Acc	***F*1**	**Acc**	***F*1**
ResNet50	**75.93**	**75.89**	**70.75**	**70.58**
Vision transformer	74.83	74.84	69.78	69.88

Note: Bold values represent the best results achieved in each metric.

performance. Overall, the models perform better without the oversampling strategy. Although oversampling yields better results than the downsampling strategy, it does not achieve the performance observed in the experiments without class balancing strategies.

Table 9 presents the training and testing times for each machine learning model under both class balancing strategies. Oversampling notably increases training time across all models, particularly for the random forest model, while testing times remain short and consistent. Nevertheless, downsampling results in shorter training times due to the reduced amount of data.

4.4.2 | Deep Learning Algorithms With Class Balancing

For this analysis, we employ the nuclei and surroundings data preprocessing strategy because it is the most promising approach, as we demonstrated in Section 4.3. We compare the deep learning models under class balancing strategies: downsampling and oversampling. The models are trained on the first fold of the PanNuke dataset and tested on the third fold. The overall metrics are presented in Table 10, where ResNet50 generally outperforms ViT.

In the per-class analysis presented in Table 11, ResNet50 achieves higher scores across all metrics, except for the downsampling strategy applied to dead cells. Some metrics improve with the oversampling strategy compared to the approach without class balancing strategies—particularly for epithelial and dead cells—although other classes exhibit slightly lower results. Consequently, the overall average performance is marginally lower than that of the model trained without oversampling. These results indicate that ResNet50 enhanced its ability to identify underrepresented classes.

Table 12 summarizes the training and testing times for deep learning models under the two class balancing strategies. As shown, ResNet50 requires less training time than the ViT in all configurations. Oversampling considerably increases training duration for both models, while testing times remain similar across strategies. Overall, ResNet50 not only achieves better performance metrics but also demonstrates greater computational efficiency.

4.5 | Comparative Analysis of DL and ML Classification Metrics

Deep learning algorithms achieve better results than machine learning algorithms, although they utilize more resources and require more time. In addition, DL algorithms do not require careful feature selection, but they do require testing and the selection of how images are presented to the algorithms. Deep learning algorithms autonomously extract the features they consider most important for cell discrimination. In contrast, machine learning algorithms perform better in terms of time and resource usage, as they do not require high-performance computing to deliver results within a reasonable time frame. However, these algorithms require much more attention to cell processing, as selecting carefully and calculating each cell feature is necessary.

Table 13 presents results for all algorithms evaluated in classification, ordered by the *F*1 metric. In addition, we include the training and test time. We select the best result for each algorithm from the various strategies tested.

Table 14 presents the comparative *F*1-scores of our best classification models (ResNet-50 and ViT) against recent state-of-the-art approaches in the PanNuke dataset. In this comparison, methods marked with an asterisk (*) focus exclusively on classification, while the remaining models perform both detection

TABLE 6 | Comparative result of classification metrics on test data for the DL algorithms for both preprocessing nuclei process by nuclei class.

Model	Nuclei class	Nuclei and surrounding		Just nuclei	
		Acc	***F*1**	**Acc**	***F*1**
ResNet50	Neoplastic	**82.68**	**82.90**	**79.37**	**78.71**
ViT		81.30	82.19	74.50	76.78
ResNet50	Connective	**72.57**	**71.49**	58.34	63.21
ViT		69.07	69.01	**66.36**	**64.08**
ResNet50	Inflammatory	70.52	**71.98**	**75.49**	**68.99**
ViT		**71.31**	70.19	70.25	68.60
ResNet50	Epithelial	72.57	71.49	**65.65**	**64.39**
ViT		**73.66**	**71.59**	64.45	63.81
ResNet50	Dead	40.68	**49.39**	37.74	41.13
ViT		**41.72**	47.93	**38.78**	**43.89**

Note: Bold values represent the best results achieved in each metric and across nuclei classes.

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TABLE 7 | Resume table for training and testing time for ResNet50 and ViT.

Model	Nuclei and surroundings			Just nuclei		
	Training (m)	Avg/epoch (m)	Test (m)	Training (m)	Avg/epoch (m)	Test (m)
ResNet50	**212**	**4.2**	**1.7**	**212**	**4.2**	**1.7**
Vision transformer	336	6.7	2.3	342	6.8	2.3

Note: Bold values represent the best time (lowest time) achieved in each stage.

TABLE 8 | Classification metrics on test data for traditional machine learning algorithms trained with raw features and class balancing strategies.

Model	Downsampling		Oversampling	
	Acc	*F*1	Acc	*F*1
Random forest	**52.70**	**54.87**	**63.81**	**62.82**
k-NN	36.77	39.82	52.63	51.06
Logistic regression	42.77	25.62	42.77	25.64
Gradient boosting	**52.70**	54.84	62.45	61.40

Note: Bold values represent the best results achieved in each metric.

TABLE 9 | Time results of classification tasks for machine learning algorithms trained on the raw features for both class balancing strategies.

Model	No. of combination of hyperparameters	Downsampling		Oversampling	
		Training (s)	Test (s)	Training (s)	Test (s)
Random forest	74	19	0.5	300	0.8
k-NN	18	**0.5**	0.4	**16**	3
Logistic regression	12	3.7	**0.004**	64	**0.004**
Gradient boosting	28	130	0.25	2514	0.29

Note: Bold values represent the best time (lowest time) achieved in each stage.

TABLE 10 | Classification metrics on test data for deep learning algorithms using both class balancing strategies.

Model	Downsampling		Oversampling	
	Acc	*F*1	Acc	*F*1
ResNet50	**66.84**	**67.5**	**75.66**	**75.75**
Vision transformer	62.11	62.83	72.86	73.04

Note: Bold values represent the best results achieved in each metric.

TABLE 11 | Comparison of classification metrics on test data for deep learning algorithms across both class balancing strategies by nuclei class.

Model	Nuclei class	Downsampling		Oversampling	
		Acc	*F*1	Acc	*F*1
ResNet50	Neoplastic	**76.78**	**76.12**	**82.56**	**82.63**
ViT		76.03	71.03	82.28	80.60
ResNet50	Connective	**60.96**	**64.23**	**72.27**	**70.98**
ViT		56.26	57.64	65.42	66.84
ResNet50	Inflammatory	**76.84**	**62.20**	**69.58**	**71.49**
ViT		56.69	60.59	67.69	68.78
ResNet50	Epithelial	**60.21**	**57.11**	**74.10**	**71.09**
ViT		57.46	53.20	70.82	69.28
ResNet50	Dead	19.43	30.56	**41.90**	**51.66**
ViT		**20.70**	**31.38**	40.75	47.14

Note: Bold values represent the best results achieved in each metric and across nuclei classes.

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TABLE 12 | Resume table for training and testing time for ResNet50 and ViT with class balancing strategies.

Model	Downsampling			Oversampling		
	Training (m)	Avg/epoch (m)	Test (m)	Training (m)	Avg/epoch (m)	Test (m)
ResNet50	**95**	**1.9**	**1.9**	**300**	**6**	**1.6**
Vision transformer	129	2.6	2.35	505	10.1	2.3

Note: Bold values represent the best time (lowest time) achieved in each stage.

TABLE 13 | Best results for accuracy, *F*1-score, and training and test time for the DL and ML algorithms.

Model	Results (%)		Time	
	Acc	*F*1	Training	Test
ResNet50	**75.93**	**75.89**	212 m	1.7 m
Vision transformer	74.83	74.84	336 m	2.3 m
Random forest	64.01	62.99	5 m	0.7 s
Gradient boosting	63.43	62.54	47 m	0.27 s
k-NN	53.87	52.69	**7.3 s**	9.7 s
Logistic regression	52.58	49.78	29 s	**0.03 s**

Note: Bold values represent the best result and the best time (lowest time) achieved across all models.

TABLE 14 | Comparison of *F*1-score (%) across nuclei classes for state-of-the-art methodologies in the PanNuke dataset.

Model	*F*1-score results (%) across nuclei classes				
	Neoplastic	Connective	Inflammatory	Epithelial	Dead
HoVer-Net [19]	62.00	49.00	54.00	56.00	31.00
CellViT-SAM-H [21]	71.00	53.00	58.00	73.00	36.00
HistoNext-Large [20]	71.00	53.00	59.00	71.00	36.00
CellNuc-DETR-Large [22]	72.00	56.00	60.00	74.00	45.00
PointNu-Net [23]	73.00	58.00	60.00	73.00	31.00
MIX-CC* [25]	88.12	61.09	66.75	58.86	45.56
ResNet50*	82.90	71.49	71.98	71.49	51.66
Vision transformer*	82.19	69.01	70.19	71.59	47.14

Note: Methods marked with an asterisk (*) focus solely on nuclei classification, while the remaining models jointly perform nuclei detection and classification.

and classification. While detection-based methods, such as CellNuc-DETR-Large and PointNu-Net, achieve strong performance across most nucleus classes, our classification-only models demonstrate superior discriminative power once nuclei have been isolated. Specifically, ResNet-50 achieves the highest *F*1-scores across connective (71.49%), inflammatory (71.98%), epithelial (71.49%), and dead (51.66%) nuclei, outperforming even specialized detection architectures in these categories. The ViT achieves comparable performance, underscoring the potential of transformer-based encoders for learning histopathological features.

These results suggest that once accurate nuclei segmentation is available, classification-focused architectures can exploit finer morphological and textural cues more effectively than current end-to-end detection frameworks. Consequently, the superior performance achieved by classification-only models indicates that existing detection + classification pipelines could be further enhanced by integrating more specialized components or techniques. This highlights the potential of hybrid approaches—where precise detection is followed by high-capacity classification networks—to achieve results comparable to, or exceeding, those of standalone classification methods, ultimately leading to more reliable and detailed cellular classification in histopathology.

4.5.1 | Comparative Analysis of DL Features' Extraction

To examine the long-range dependencies captured by the ViT model, we analyze the features extracted by both deep learning models using the Grad-CAM [32]. Grad-CAM is chosen because it provides an intuitive, spatially interpretable visualization of the regions that most influence the model's decisions, enabling a direct comparison of the attention focus between architectures. Moreover, prior work by Azmoodeh-Kalati et al. [33] demonstrated and analyzed deep learning features extraction using Grad-CAM as a visualization tool. Figure 9 shows Grad-CAM visualizations comparing the attention patterns of ViT and ResNet50 models on representative neoplastic cell samples. The first two columns show the original and preprocessed image patches used for model input. The third and fourth columns exhibit the Grad-CAM heatmaps generated by each model, highlighting that

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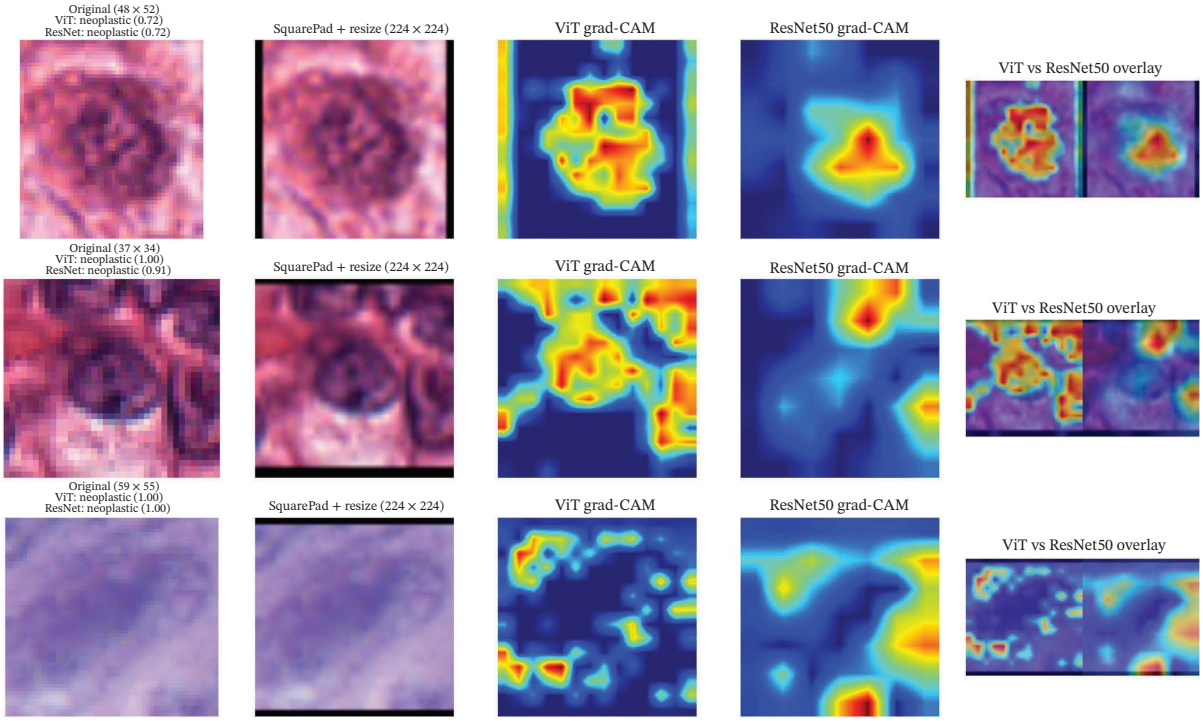

FIGURE 9 | Grad-CAM visualizations for ViT and ResNet50 models on neoplastic cell samples. Each row shows the original cropped patch, the preprocessed (SquarePad + resized) version, the Grad-CAM activation maps for ViT and ResNet50, and their overlay comparison. Both models focus on the nucleus region, though ViT exhibits broader activation patterns, while ResNet50 produces more localized responses around the nuclear structure.

the regions are most influential in their classification decisions. The fifth and final column presents a comparative view of the Grad-CAM heatmaps across models. ResNet50 primarily concentrates its attention on compact, nucleus-centered regions, indicating a focus on key morphological structures. In contrast, the ViT exhibits more diffuse activation across the cell and its surroundings, suggesting a broader contextual interpretation. The overlay comparison demonstrates that, despite differences in focus patterns, both models identify the nucleus as the main discriminative region for neoplastic classification.

5 | Conclusion

In this article, we study the behavior of several classification models, including traditional machine learning algorithms and deep learning algorithms. We train all the algorithms tested by using different preprocessing strategies. The main objective is to gain insight into the various strategies employed by the CPath community for nuclei classification and to systematically compare the effectiveness of different classification algorithms in this context. Regarding ML algorithms, we evaluate random forest, gradient boosting, k-NN, and logistic regression. Regarding DL models, we evaluate ResNet50 and ViT.

Results demonstrate the potential of ResNet50 and ViT as backbone architectures for nuclei classification tasks. Both deep learning methods outperform traditional ML approaches, highlighting the limitations of the ML algorithms. Traditional ML algorithms rely on feature engineering, selection, and calculation. In contrast, deep learning algorithms bypass this constraint by learning features automatically during training. This mechanism allows them to adapt more effectively to complex datasets. However, this advantage comes with increased computational demands, as DL models require significantly more time and resources to train and evaluate than traditional ML methods.

Among the deep learning algorithms compared, ResNet50 consistently delivers slightly better results across both preprocessing strategies, showing its robustness and efficiency. However, ViTs demonstrate a notable advantage in classifying underrepresented nuclei classes, potentially due to their capacity for capturing long-range dependencies and relationships within the data. Both models underscored the importance of cell context in classification tasks, including the surrounding area of the cell as part of the input improved classification performance, suggesting that the local cellular environment provides critical information for distinguishing between nuclear types. Furthermore, our experiments incorporating class balancing strategies provided additional insights into handling the inherent data imbalance of the PanNuke dataset. Both downsampling and oversampling approaches are evaluated, with oversampling through data augmentation leading to improved recognition of underrepresented nuclei classes such as dead cells. These results highlight the importance of addressing class imbalance when developing robust nuclei classification models.

Some potential future directions include exploring improved classification algorithms and refining those presented in this study. By investigating advanced model architectures or optimization techniques, it may be possible to further enhance classification accuracy and robustness. This could include experimenting with hybrid models that combine the strengths of traditional machine learning, CNNs, and transformers, or

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integrating cutting-edge innovations in deep learning to push the boundaries of current methodologies, like an improved version of ViT with registers developed by Darcet et al. [34].

Another avenue could involve incorporating additional datasets to enhance the robustness of the models. Previous studies have demonstrated how combining multiple datasets can improve generalization. For example, Parulekar et al. [35] unify three datasets and their labels, proposing a novel unified hierarchical classification system. Similarly, Zhan et al. [36] introduce correlation-based cross-training to improve model performance across datasets. Another option could be optimizing the use of cell contexts, which, as we have seen here, differ in the nuclei classification.

A future direction could explore stain normalization and data augmentation techniques, such as those proposed by Pérez-Bueno et al. [37] and Shen et al. [38]. Combining these techniques with the integration of multiple datasets could address the generalization issues often encountered in histopathology models. Additionally, this approach could mitigate the underrepresentation of the nuclei class in the PanNuke dataset, leading to more reliable and widely applicable results.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are openly available in the PanNuke Dataset–Tissue Image Analytics (TIA) Centre at https://warwick.ac.uk/fac/cross_fac/tia/data/pannuke.

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